

Also see Daisy Chain *

□ Nested Interrupt:

Example: Suppose in ~~in~~ CPU 2, 4 peripheral devices (A, B, C & D) are connected. Their respective priority is 1, 2, 3, 4 (High = 1). State the following procedure

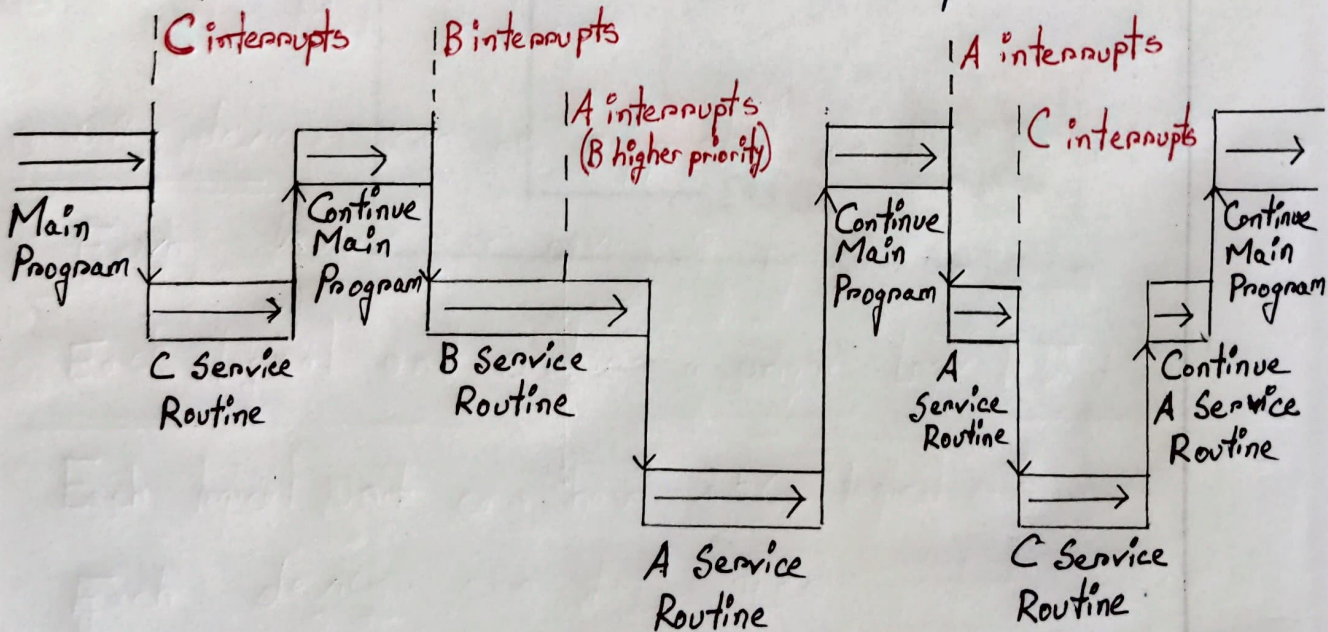
- ① Device C interrupts main program and its service routine runs to complete

- ② Device B interrupts main program, in turn interrupted by A.

After that both service routine runs to complete

- ③ Device A interrupts main program, in turn interrupted by C.

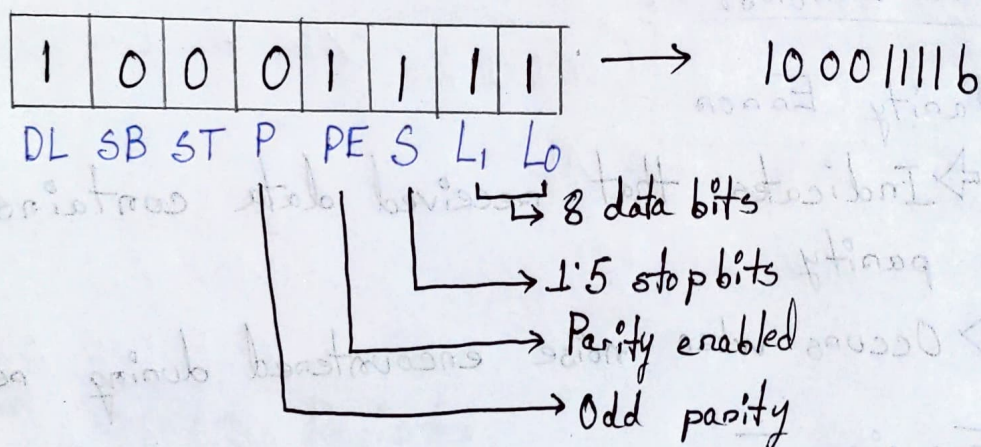
After that both service routine runs to complete



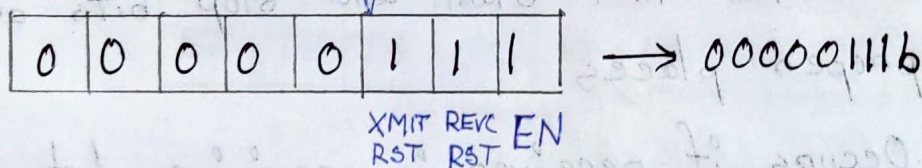
Example

Suppose an asynchronous system requires 8 data bits, odd parity, baud rate of 1200 and 1.5 stop bits. The i/o port is A0h and A1h. Write a program to initialize the K550 UART

① Line Control Register



② FIFO Control Register



③ Programming Baud Rate

$$\frac{18.432 \times 10^6}{16 \times x} = 1200$$

∴ Divisor, $x = 960 \rightarrow 0000001111000000b$

Sent to
MSB

Sent to LSB

④ Port Addresses

Baud Rate Divisor → A0h (LSB) & A1h

FIFO Control Register → A2h

Line Control Register → A3h

} Maintain this sequence

Code

LSB EQU A0h

MSB EQU A1h

FIFO EQU A2h

LINE EQU A3h

INIT PROC NEAR

MOV AL, 10001111b

OUT LINE, AL

MOV AL, 11000000b

OUT LSB, AL

MOV AL, 00000011b

OUT MSB, AL

MOV AL, 00001111b

OUT LINE, AL

MOV AL, 00000111b

OUT FIFO, AL

RET

INIT ENDP

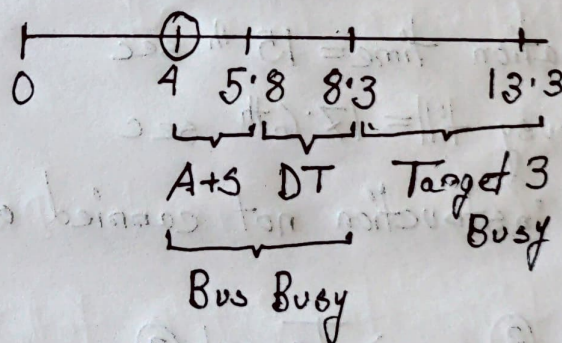
CT-2 Question

Initiator	Target	Arbitration time of Initiator (sec)	Task time of Target given by Initiator (sec)
3	3	4	5
1	3	7.2	10
2	3	11	8.4
2	2	15	12.6
3	2	18	5

Data Transfer time = 2.5 sec

Arbitration + Selection / Reselection time = 1.8 sec

1) Initiator ③ → Target ③



2) Initiator ① → Target ③

Arbitration time = 7.2th second

Bus Busy till = 8.3th second

∴ 2nd Instruction NOT carried out

3) Initiator ② → Target ③

... 11 12.8 13.3

A+S Target 3

Bus Busy

2

Target 3 Queue

Next instruction's arbitration at 15th s. in given scenario. However, Queued instruction can be executed at 13.3 s. ∴ Queued instruction executed first

4) Initiator ② → Target ③

... 13.3 15.1 17.6 20.6

A+S DT Target 3

Bus Busy

Target 3 Queue

5) Initiator ② → Target ②

Arbitration time = 15th sec

Bus busy till = 17.6th sec

∴ 4th instruction not carried out

6) Initiator ③ → Target ②

... 18 19.8 22.3 27.3

A+S DT Target 2

Bus Busy

Bus free after 22.3 seconds

All controllers free after 27.3 seconds

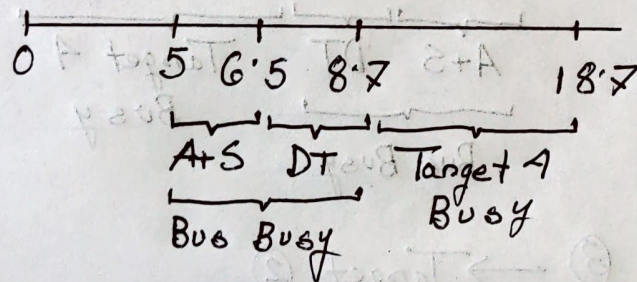
SCSI Math (Prev Year Question)

Initiator	Target	Arbitration Time of Initiator (sec)	Task Time of Target given by Initiator (sec)
2	4	5	10
1	2	8.1	20.5
2	4	16	5.8
1	4	16.5	3
3	2	21	12
3	3	24	2.6

Data Transfer time = 2.2 sec

Arbitration + Selection/Reselection Time = 1.5 sec

1) Initiator ② → Target ④



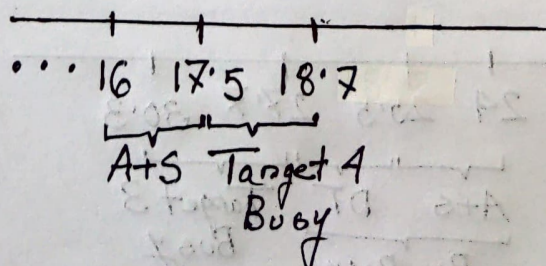
2) Initiator ① → Target ②

Arbitration time = 8.1 sec

Bus busy till 8.7 sec

∴ 2nd Instruction not executed

3) Initiator ② → Target ④



2	
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Target 4 Queue

4) Initiator ① → Target ④

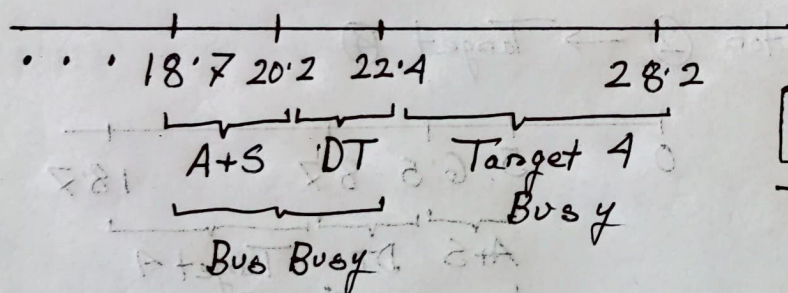
Arbitration time = 16.5 s

Bus busy till = 17.5 s

∴ 4th Instruction not executed

5) Initiator ② → Target ④

Next instruction's arbitration time is 21 sec in given scenario. But Queued instruction can be executed at 18.7 sec, ∴ Queued instruction executed first



Target 4 Queue

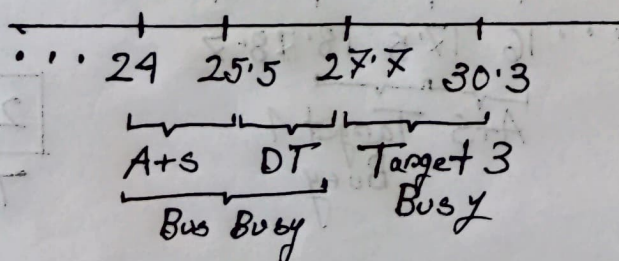
6) Initiator ③ → Target ②

Arbitration time = 21 sec

Bus busy till 22.4 sec

5th Instruction not executed

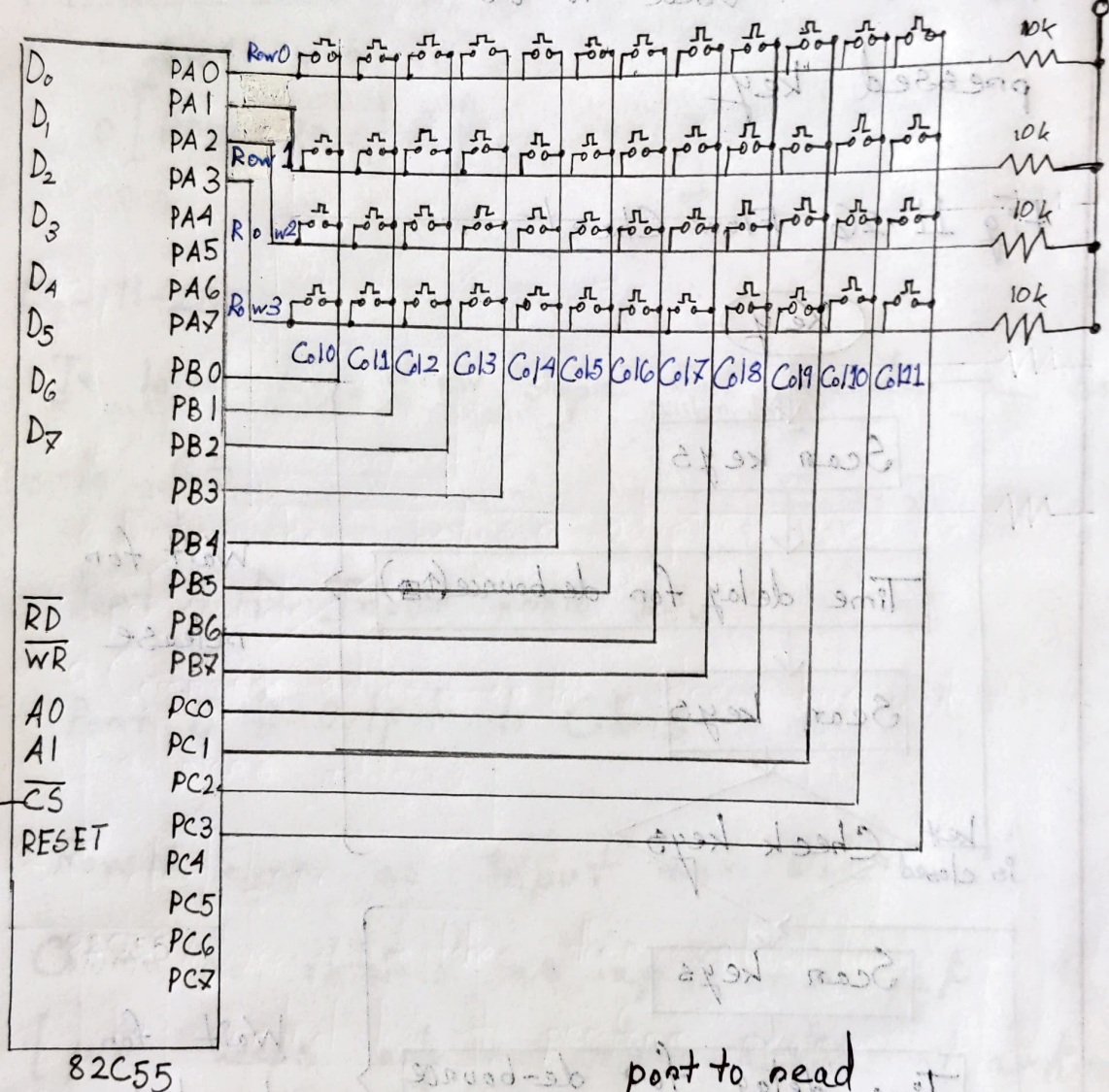
7) Initiator ③ → Target ③



Bus free after 27.7 sec

All controllers free after 30.3 sec

4x12 Keyboard



Port A (PA0-PA3) used as input from Rows

Port B (PB0-PB7) and Port C (PC0-PC3) used as output port - to select columns

Command Byte of 82C55

1	0	0	1	1	0	0	0
---	---	---	---	---	---	---	---

Group A mode 0
Port A used as input
Port C (PC0-PC3) used as input (Don't Care)

(PC0-PC3)
Port C as Output
Port B as Output
Group B mode 0

- One column is selected at a time by 82C55, and the selected column shifts very fast.
- If 1111 is output to Port C and 11111101 to Port B the four keys in Column 1 can be selected.
- When user presses a key, the row number is obtained and the column number is obtained from the column selected at that time, and thus correct key is identified

Other Matrices

- $2 \times 8 \rightarrow$ Port A (PA0 and PA1) for input from rows
Port B (PB0-PB7) for output to select columns
- $8 \times 16 \rightarrow$ Port A (PA0-PA7) for input from rows
Port B (PB0-PB7) and Port C (PC0-PC7) to select columns
- $12 \times 12 \rightarrow$ Port A (PA0-PA7) and Port C (PC4-PC7) as input ports to read from rows
Port B (PB0-PB7) and Port C (PC0-PC3) as output ports to select columns

Notes: Port C can be used half for input and half for output

Barcode Checksum

Example-2

243160102439X

Consists of 13 digits $\therefore$ EAN

[No leading
0 required]

o $243160-102439-X$

o $4+1+0+0+4+9 = 18$

o $18 \times 3 = 54$

o $2+3+6+1+2+3 = 17$

o $54+17 = 71$

o $71+X = 80$

o $\therefore X = 9$